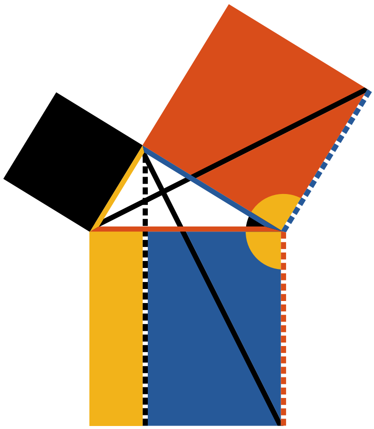


Book I. Prop. XLVII. Theor. *In a right angled triangle*  *the square on the hypotenuse is equal to the sum of the squares of the sides (and).*

On , , describe squares (pr. I.46)

Draw ‖ (pr. I.31); also draw and .

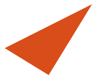
 =  .

To each add  : ∴  =  ,

= and = ;

∴  =  (pr. I.4).

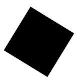
Again ∵ ‖

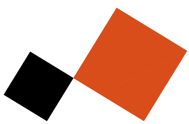
 = twice  (pr. I.41),

and  = twice  (pr. I.41);

∴  =  .

In the same manner it may be shown that

 =  ;

hence  =  .

Q. E. D.